

From Words to Market Moves: Comparing Rule-Based, Statistical, and Transformer Models for Financial Language

From Neurons to Transformers: Cognitive Principles in AI Research
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Abstract

Financial sentiment classification provides a useful setting for studying how different representations of language affect model behavior. This project compares three paradigms: a rule-based system using the Loughran–McDonald financial dictionary, a statistical TF–IDF representation with logistic regression, and a finance-domain BERT model fine-tuned on Financial PhraseBank. After removing five exact duplicates, 3,448 labeled financial sentences are divided into a single frozen train–validation–test split, with 518 sentences reserved for held-out evaluation. Accuracy increases from 64.86% for the rule-based system to 81.85% for TF–IDF and 93.24% for Finance BERT; macro-F1 increases from 0.484 to 0.734 and 0.913, respectively. Beyond aggregate performance, the models exhibit distinct error patterns. The simpler representations strongly favor the neutral class, whereas Finance BERT achieves balanced recall across all three classes. Analysis of all 35 remaining Finance BERT errors and 24 controlled diagnostic probes nevertheless reveals structured failures when sentiment depends on the interaction between a financial metric and its direction of change. The results suggest that contextual representations substantially improve financial sentiment classification without guaranteeing systematic domain-specific reasoning.

1 Introduction

Financial sentiment cannot always be inferred from individual positive or negative words. Consider two nearly identical statements:

Operating profit increased from EUR 10 million to EUR 15 million.

Operating loss increased from EUR 10 million to EUR 15 million.

The word *increased* appears in both sentences, yet the first describes an improvement while the

second describes a deterioration. Correct interpretation therefore depends on the relationship between a directional expression and the financial quantity to which it applies.

This makes financial language a useful domain for comparing different approaches to representation. Finance-specific dictionaries such as Loughran and McDonald (Loughran and McDonald, 2011) encode explicit human-defined lexical categories. Statistical approaches such as TF–IDF represent text through weighted lexical features (Spärck Jones, 1972), allowing a classifier to learn associations from labeled examples. Transformer models such as BERT construct contextual representations in which token meaning depends on surrounding language (Devlin et al., 2019). Finance-domain BERT models extend this idea through pre-training on financial communications (Yang et al., 2020).

This progression mirrors a broader seminar theme: moving from explicit rule-like representations toward example-driven and distributed neural representations. The central research question is therefore:

How do rule-based, statistical, and transformer-based models differ in the way they represent and interpret financial language?

Rather than treating the task only as a model-ranking exercise, the analysis examines overall performance, class-specific errors, linguistic subsets, disagreement patterns, qualitative failure categories, and controlled diagnostic probes.

Analytical expectations. The rule-based representation should depend strongly on lexical coverage because its decision is determined by dictionary matches. TF–IDF can learn broader unigram and bigram associations from data, but it does not construct context-dependent token representations.

Sentence	Label
Profit for the period was EUR 9.8 mn, up from EUR 6.0 mn in 2004.	Positive
Operating profit fell to EUR 20.3 mn from EUR 74.2 mn.	Negative

Table 1: Representative Financial PhraseBank observations.

Finance BERT should provide stronger contextual discrimination; whether contextualization is sufficient for systematic composition of financial metric identity, direction, temporality, and competing cues remains an empirical question.

2 Related Work

Financial sentiment analysis differs from generic sentiment classification because terms that appear negative in ordinary language can have context-dependent interpretations in finance. Loughran and McDonald (Loughran and McDonald, 2011) showed that general-purpose sentiment dictionaries can mischaracterize financial disclosures and introduced finance-specific lexical categories. Financial PhraseBank later provided a human-annotated benchmark of financial news sentences (Malo et al., 2014).

Sparse lexical representations remain a useful statistical baseline because they directly expose the value of learned word and phrase associations. TF-IDF traces to statistical term specificity in information retrieval (Spärck Jones, 1972). In contrast, BERT (Devlin et al., 2019) represents tokens contextually through bidirectional transformer layers. The finance-domain model used here follows Yang et al. (Yang et al., 2020), who pretrained BERT on financial communications.

3 Data and Experimental Setup

3.1 Financial PhraseBank

The experiments use the sentences_75agree configuration of Financial PhraseBank (Malo et al., 2014). Each observation consists of one financial sentence and one of three labels: negative, neutral, or positive.

Two representative observations illustrate the data structure:

The original corpus configuration contains 3,453 observations. Five exact duplicates are removed, leaving 3,448 observations: 2,141 neutral (62.09%), 887 positive (25.73%), and 420 nega-

tive (12.18%). A single stratified split with random seed 42 is frozen for all experiments: 2,413 training sentences, 517 validation sentences, and 518 held-out test sentences.

3.2 Evaluation

All three models are evaluated on the same frozen 518-sentence test set. The primary aggregate metrics are accuracy and macro-F1. Macro-F1 is important because the dataset is class-imbalanced and therefore weights the three sentiment classes equally. We also examine per-class recall, confusion directions, model disagreements, and behavior on controlled diagnostic probes.

4 Models

4.1 Rule-Based Representation

The first model uses the Loughran–McDonald financial sentiment dictionary (Loughran and McDonald, 2011). Tokens are matched against positive and negative word lists. A simple three-token backward window handles local negation. Let N_+ and N_- denote the resulting positive and negative counts:

$$s = N_+ - N_-.$$

A positive score yields a positive prediction, a negative score a negative prediction, and $s = 0$ a neutral prediction. The model does not learn from PhraseBank.

4.2 TF-IDF and Logistic Regression

The statistical model represents each sentence using TF-IDF weighted unigrams and bigrams. The training set produces a sparse matrix of 2,413 sentences and 7,969 lexical features. Multiclass logistic regression learns weights relating these features to the three labels. Unlike the rule system, the polarity associated with words and short phrases is learned from the labeled training examples rather than supplied manually.

4.3 Finance BERT

BERT introduced deep bidirectional contextual representations (Devlin et al., 2019). We use the finance-domain pretrained model of Yang et al. (2020) with a three-class classification head. The full model is fine-tuned on the frozen training split for three epochs with learning rate 2×10^{-5} , batch size 8, weight decay 0.01, and maximum sequence length 128. The checkpoint with the highest validation macro-F1 is selected before test evaluation.

Model	Accuracy	Macro-F1
Rule-based	0.649	0.484
TF-IDF + LR	0.819	0.734
Finance BERT	0.932	0.913

Table 2: Held-out Financial PhraseBank results.

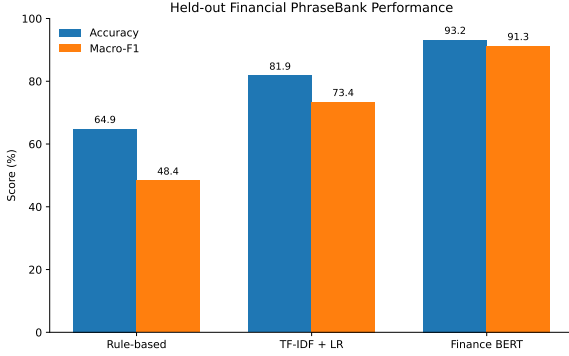


Figure 1: Held-out accuracy and macro-F1 across the three representation paradigms.

5 Results

5.1 Overall Benchmark Performance

Table 2 and Figure 1 show a clear increase in performance across the three representation paradigms.

The macro-F1 improvement is especially important because it shows that Finance BERT’s advantage is not limited to the majority neutral class.

5.2 Class-Specific Behavior

The aggregate improvement masks substantial differences in error structure. The rule-based model recalls 89.8% of neutral sentences but only 33.3% of negative and 19.5% of positive sentences. TF-IDF exhibits a similar, although weaker, pattern: neutral recall reaches 98.4%, compared with 47.6% for negative and 57.9% for positive examples. Finance BERT instead achieves recall between 90.5% and 94.4% across all three classes.

5.3 Linguistic Subsets and Disagreement

Exploratory subsets were identified for negation, contrast, expectation, and uncertainty. Finance BERT performed strongly on the observed negation and contrast subsets, although the subset sizes are small (13–24 sentences) and should not be treated as population-level estimates.

A disagreement analysis provides a complementary view. All three models are simultaneously correct on 285 of 518 test sentences. Finance BERT

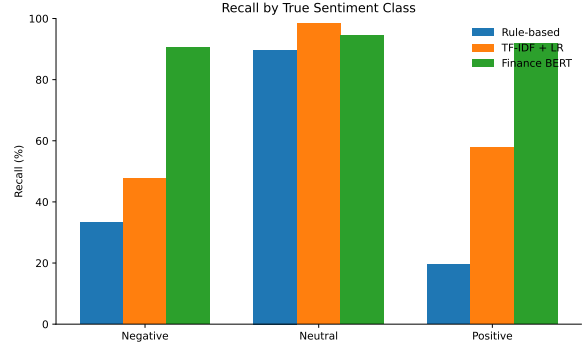


Figure 2: Recall by true sentiment class.

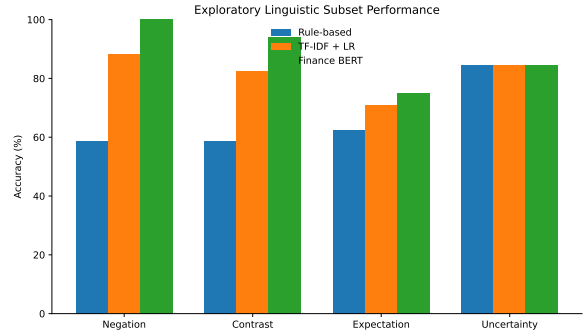


Figure 3: Exploratory accuracy on linguistic subsets.

and TF-IDF are jointly correct on another 118, while Finance BERT alone is correct on 46.

6 Error Analysis

Finance BERT misclassifies only 35 of the 518 held-out sentences. Manual coding of all 35 errors reveals that the remaining failures are not homogeneous.

The largest category contains 11 neutral-boundary or lexical overinterpretation errors. Eight cases involve implicit business-event consequences, seven involve future/expectation/realization distinctions, and seven involve comparative financial reasoning. Contrast composition and recommendation/benchmark semantics account for one error each.

Several natural test errors are particularly informative:

These examples share a common property: sentiment cannot be inferred from the directional term alone. An increase in profit is favorable, whereas an increase in loss is unfavorable; a decrease in losses can be positive despite the negative lexical item *loss*.

Sentence	Gold	BERT
Diluted loss per share stood at EUR 0.15 versus EUR 0.26.	Positive	Negative
Return on investment was 4.1% compared with 43.8%.	Negative	Positive
Operating loss increased to EUR 17 mn from EUR 10.8 mn.	Negative	Positive

Table 3: Representative comparative Finance BERT errors from the held-out PhraseBank test set.

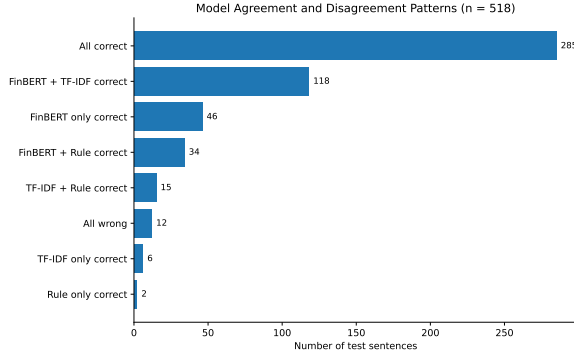


Figure 4: Agreement and disagreement patterns across the 518 held-out test sentences.

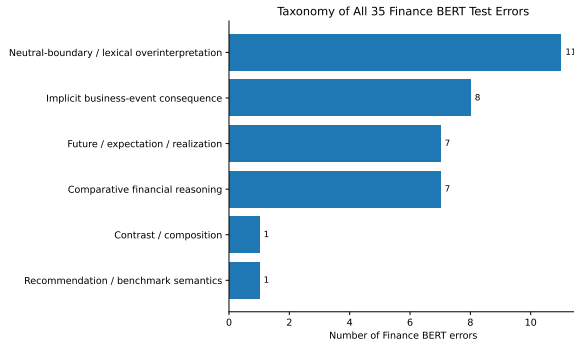


Figure 5: Interpretive taxonomy of all 35 Finance BERT test errors.

7 Controlled Diagnostic Probes

The qualitative analysis suggested recurring failure modes but could not establish whether they were reproducible under controlled linguistic changes. We therefore constructed and froze 24 diagnostic sentences before model inference. These probes manipulate comparative financial reasoning, negation, intention versus realization, contrast, and uncertainty. Their labels are researcher-defined semantic targets and are therefore reported as *target alignment*, not as a second benchmark accuracy measure.

The clearest pattern occurs in metric–direction composition:

Finance BERT aligns with the intended target on all profit and revenue direction probes but fails on all four analogous loss and cost inversions. The pat-

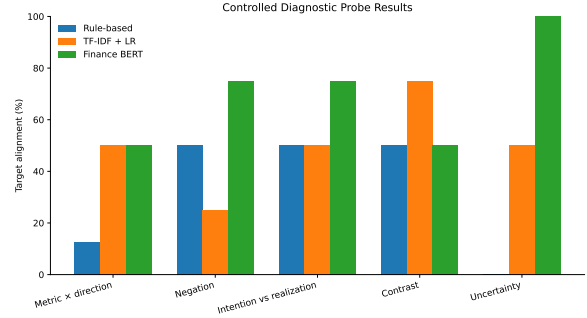


Figure 6: Target alignment on the 24 controlled diagnostic probes.

tern is structured rather than uniformly distributed across the eight sentences. It is consistent with strong sensitivity to directional terms such as *increased* and *decreased* without reliable composition of that direction with the economic polarity of the underlying metric.

Other probe families reveal more nuanced behavior. Finance BERT correctly distinguishes *achieved profitability* from *did not achieve profitability*, but predicts neutral for *did not secure a major contract*. It distinguishes *considering entering a new market* from a realized market entry, yet interprets *plans to expand production capacity* as positive rather than neutral. In contrastive probes, both mixed-signal examples receive negative predictions under a predefined neutral target. On four uncertainty probes, replacing *will* with *may* preserves the intended polarity but changes model confidence only minimally.

8 Discussion

The three systems illustrate a progression from explicit lexical rules to learned lexical associations and finally contextual representations. The rule-based model is transparent but strongly constrained by vocabulary coverage: only about 18% of test sentences contain usable positive or negative lexicon evidence, leading frequently to neutral predictions. TF-IDF learns a much broader set of lexical patterns and improves substantially, but it still exhibits a strong neutral tendency.

Controlled sentence	Target	BERT	Match
Operating profit increased from EUR 10m to EUR 15m.	Positive	Positive	✓
Operating profit decreased from EUR 15m to EUR 10m.	Negative	Negative	✓
Revenue increased from EUR 10m to EUR 15m.	Positive	Positive	✓
Revenue decreased from EUR 15m to EUR 10m.	Negative	Negative	✓
Operating loss increased from EUR 10m to EUR 15m.	Negative	Positive	×
Operating loss decreased from EUR 15m to EUR 10m.	Positive	Negative	×
Operating costs increased from EUR 10m to EUR 15m.	Negative	Positive	×
Operating costs decreased from EUR 15m to EUR 10m.	Positive	Negative	×

Table 4: Controlled metric–direction probes for Finance BERT. Targets were defined before inference.

Finance BERT removes much of this imbalance and achieves much stronger contextual classification. Yet the controlled probes show that contextual representation does not necessarily imply systematic composition of domain-specific relations. In the metric–direction probes, sentiment is effectively a function of both the metric and its movement:

$$\text{sentiment} = f(\text{metric}, \text{direction}).$$

Finance BERT handles that relation reliably for profit and revenue in this diagnostic set, but not for losses and costs.

This result is relevant to the seminar’s rule-versus-example framing. The rule system contains explicit symbolic decisions; TF–IDF learns surface statistical associations over observed features; Finance BERT compresses regularities into distributed parameters and contextual representations. Its superior benchmark performance therefore does not imply that it has acquired an explicit or fully systematic financial rule. The behavior observed here is more consistent with powerful learned generalization that can still break under controlled compositional changes.

Limitations

The linguistic subsets and controlled diagnostic suite are small and should be interpreted as exploratory behavioral stress tests, not population-level estimates of model capability. The 24 synthetic probe targets are researcher-defined rather than independently annotated gold labels. Manual error categories involve interpretive judgment, and the rule-based baseline is deliberately simple rather than representative of every possible symbolic financial NLP system. Finally, the transformer examined here is a BERT-era finance model rather than a frontier instruction-tuned language model.

9 Future Work

The same framework could be extended to modern instruction-tuned language models and substantially harder financial text such as earnings calls, analyst commentary, and multi-sentence disclosures. A multilingual extension could test parallel financial constructions in English, German, Spanish, and French to determine whether metric–direction reasoning, negation, temporality, and contrast transfer across languages or depend strongly on language-specific lexical cues.

10 Conclusion

Increasing representational sophistication produces large gains in financial sentiment classification: accuracy rises from 64.86% for explicit lexical rules to 81.85% for learned TF–IDF features and 93.24% for Finance BERT. More importantly, the nature of model errors changes. Rules are constrained by lexical coverage, TF–IDF learns broader but still strongly lexical associations, and Finance BERT provides much more balanced contextual discrimination.

However, strong benchmark performance does not guarantee systematic financial reasoning. Natural errors and controlled diagnostics reveal failures when sentiment depends on the interaction between metric identity and direction of change. The results therefore support a distinction between increasingly powerful contextual language representation and robust domain-specific compositional reasoning.

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